



Butyl rubber-based interpenetrating polymer networks with side chain crystallinity: Self-healing and shape-memory polymers with tunable thermal and mechanical properties

Esra Su^a, Gozde Bayazit^b, Semra Ide^b, Oguz Okay^{a,*}

^a Department of Chemistry, Istanbul Technical University, 34469 Maslak, Istanbul, Turkey

^b Department of Physics Engineering and Department of Nanotechnology and Nanomedicine, Hacettepe University, 06800 Beytepe, Ankara, Turkey

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ABSTRACT

A self-healing and shape-memory interpenetrating polymer network (IPN) is produced by UV polymerization of n-octadecyl acrylate (C18A) monomer in a toluene solution of butyl rubber (isobutylene-isoprene rubber, IIR) using Irgacure 2959 photoinitiator at ambient temperature. IPNs containing 20–80 wt% IIR have crystalline domains formed by side-by-side packed octadecyl side chains aligned perpendicular to the poly(C18A) (PC18A) backbone. TEM images reveal that the morphology of IPNs consists of crystalline domains dispersed in a continuous amorphous matrix where the size of the dispersed phase could be adjusted between μm and nm level by changing the amount of IIR. Calculations indicate that the effective cross-link density of IPNs is mainly determined by their crystalline domains followed by hydrophobic associations while the chemical cross-links between IIR and PC18A components are negligible. We also show that the crystalline domains acting as sacrificial bonds dissipate energy under strain leading to a significant toughness improvement. IPNs exhibit tunable melting temperature (46–50 °C), crystallinity (1.5–25%), Young's modulus (0.6–35 MPa), toughness (1.3–12 MPa), and a stretchability of up to 1200% by varying the amount of IIR component. What is more, they also exhibit temperature induced healing behavior with 59–77% efficiency, and an effective shape-memory function. The strategy presented here is applicable for the preparation of IPNs based on various rubbers and poly(n-alkyl (meth)acrylates) with long alkyl side chains.

1. Introduction

Elastomers or rubbers are soft polymer networks characterized by their very high stretchability and complete recoverability. These characteristics originate from their flexible and long polymer chains with entropic elasticity, that are joined together into a 3D network structure, usually via covalent sulfur cross-links [1]. One would expect that the replacement of the permanent sulfur cross-links with non-covalent ones that can break and reform reversibly would result in a dynamic elastomeric network with smart functions such as the self-healing ability. In recent years, preparation of physically cross-linked self-healable commercial rubbers with good mechanical properties attracts significant interest to prolong their lifetimes and to reduce waste generation and disposal costs [2]. Moreover, shape-memory is another smart function of elastomers for their various biomedical applications including sensors, flexible electronics, implants, actuators, and smart medical devices [3].

To create self-healing in commercial rubbers, sulfur vulcanization was carried out by adding copper catalysts to trigger disulfide metathesis reaction in vulcanized rubber and hence, to create reversible cross-links [4,5]. By manipulating vulcanization conditions, Hernandez et al. increased the fraction of dynamic disulfide bonds in rubber without modifying the commercial rubber formulation [6,7]. Alternatively, dynamic covalent bonds, hydrogen bonding and ionic interactions were utilized instead of sulfur vulcanization to create reversible cross-links in commercial rubbers [2,8–23]. For instance, functionalization of bromobutyl rubber (BIIR) with butylimidazole creates pendant imidazolium bromide groups on BIIR that are able to form ionic associations serving as reversible cross-links, and providing formation of rubbers with 1000% stretchability and around 9 MPa tensile strength [18]. It was shown that the tensile strength of the ionically cross-linked rubbers decreases while their self-healing efficiency increases as the alkyl chain length of alkyl imidazoles increases [19]. Alternatively, rubber-based

* Corresponding author.

E-mail address: okay@itu.edu.tr (O. Okay).

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